

Random Leakages

September 28, 2026

Overview

The runs collected here all use the single-influx-resource minimal MiCRM model (`scripts/single_influx.jl`): a well-mixed community is sampled, its steady state is found by integrating the ODEs, and the outcome is classified as *extinct* (all consumers die out), *stable* (a well-mixed coexistence steady state that also survives a linear stability check against spatial perturbations at a range of wavenumbers), or *unstable* (the steady state is only stable when well-mixed, and a spatial instability is found). This is repeated many times over a grid of the normalized energy supply rate K and the shape of the per-strain leakage fraction, which is drawn from a Beta distribution for each sampled community. The two runs below differ only in how that Beta distribution is parametrized and swept.

1 `single_influx_randl2` / `main2`

Data. Directory: `cluster_env/runs/single_influx_randl2`, generated by `main2()` in that directory's `base.jl` (submitted via `job2.slurm`).

Description. This run parametrizes the leakage Beta distribution by a target *mean* and a *spread factor*: the spread factor is the standard deviation expressed as a fraction of the largest value that keeps the distribution unimodal for that mean (so a factor near 0 is a near-deterministic leakage value, and 1 is the widest unimodal spread possible). The mean is swept over 15 values, non-linearly rescaled to concentrate samples near a leakage of 1 (the biologically relevant high-leakage regime), and the spread factor is swept over 4 values, {0.0009, 0.045, 0.156, 0.333}, from essentially fixed leakage up to a third of the maximum non-bimodal spread. K is swept over 30 log-spaced values from 1 to ~ 3160 . With 150 repeats per (K , mean, spread) combination this is 270,000 individual runs.

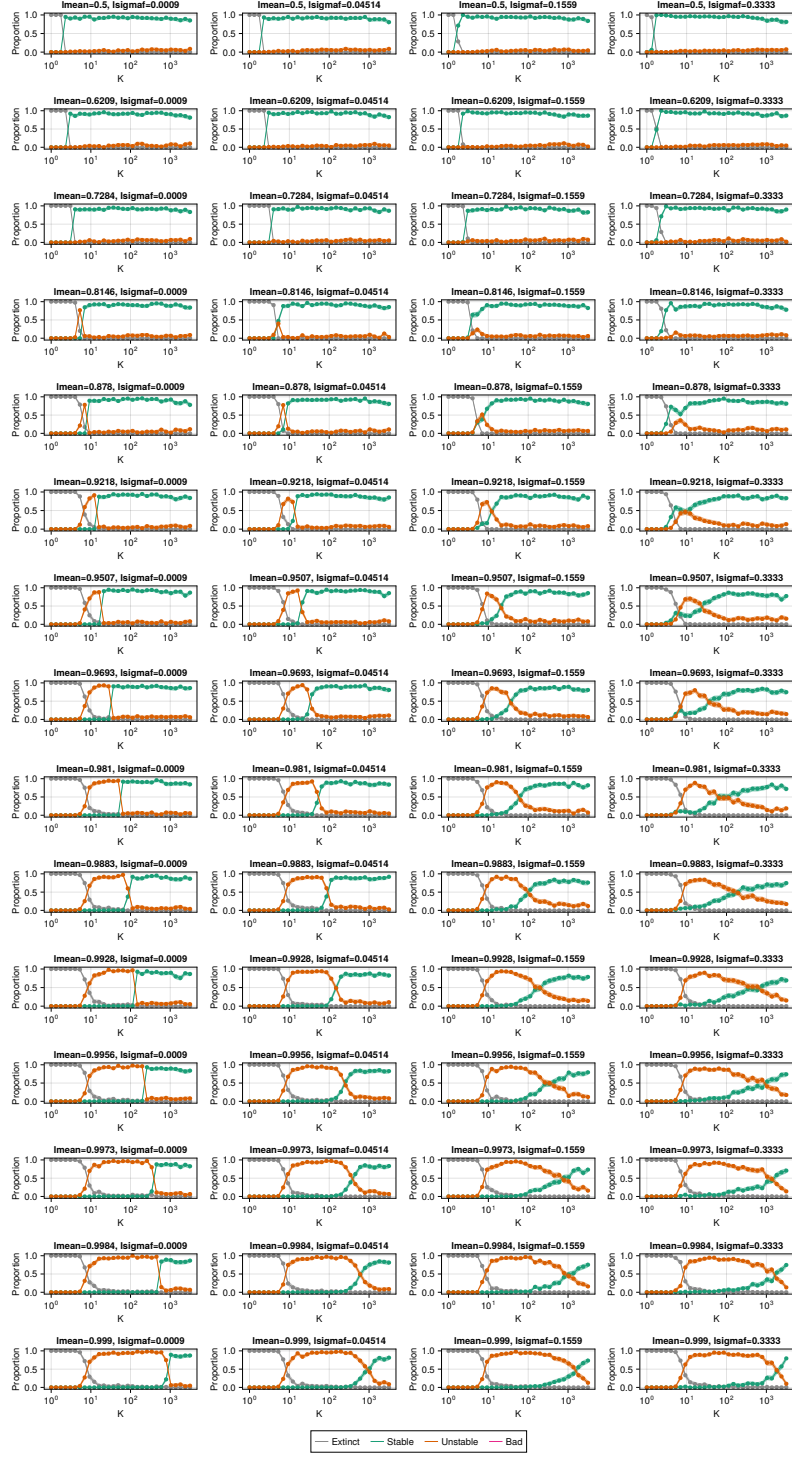


Figure 1: Proportion of extinct/stable/unstable outcomes vs. K , tiled by mean leakage (rows) and spread factor (columns).

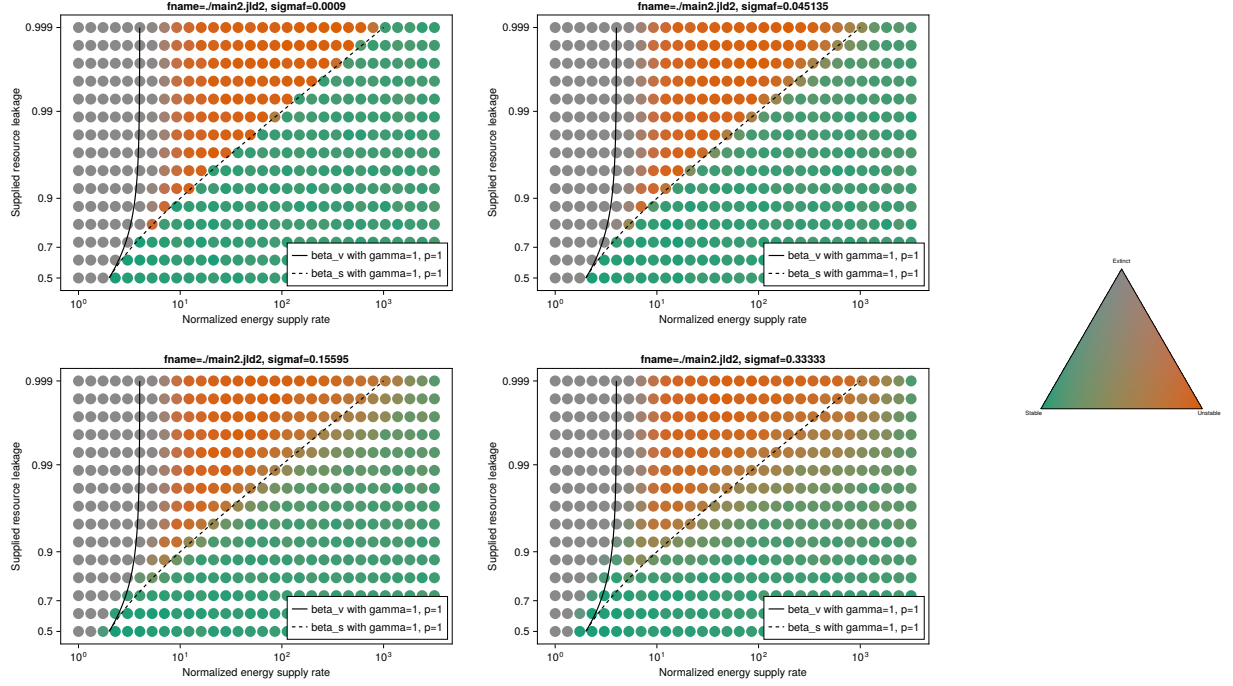


Figure 2: Outcome phase diagram (K vs. leakage, coloured by the ternary extinct/stable/unstable proportion), one panel per spread factor index (1 to 4, narrowest to widest), with the colour legend on the right.

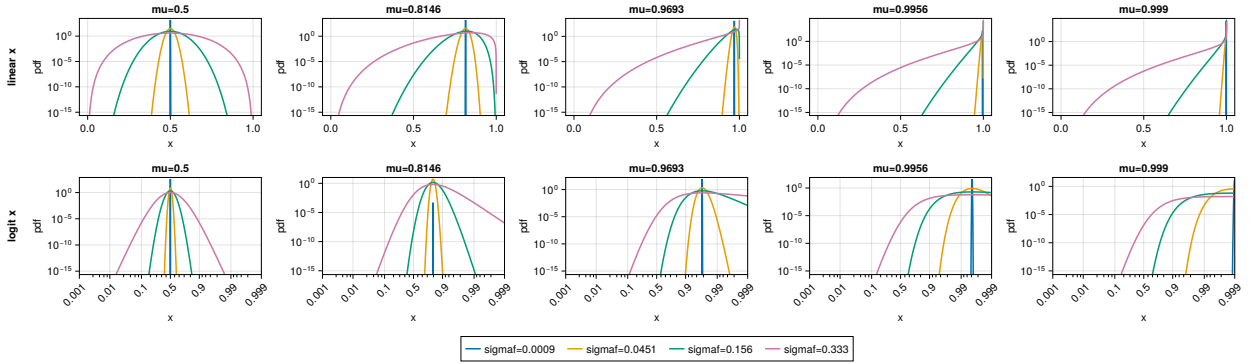


Figure 3: The sampled leakage distribution, $\text{Beta}(\mu, \sigma_f \cdot \sigma_{\text{max}}(\mu))$, for five representative mean-leakage values spread across the swept range (including the smallest and largest), each panel showing all four spread factors σ_f .

2 single_influx_randl3 / main1

Data. Directory: `cluster_env/runs/single_influx_randl3`, generated by `main1()` in that directory's `base.jl` (submitted via `job1.slurm`).

Description. This run instead parametrizes the leakage Beta distribution directly by its native shape parameters α and β , each swept over the same 6 dyadic values $\{0.5, 1.32, 3.48, 9.19, 24.25, 64\}$ (all 36 combinations), which covers everything from U-shaped/bimodal distributions ($\alpha, \beta < 1$) through to sharply peaked, nearly deterministic ones ($\alpha, \beta \gg 1$), including both symmetric ($\alpha = \beta$, centred at a leakage of 0.5) and skewed combinations biased towards low or high leakage. K is swept over 50 log-spaced values from 1 to ~ 3160 . With 150 repeats per (K, α, β) combination this is also 270,000 individual runs.

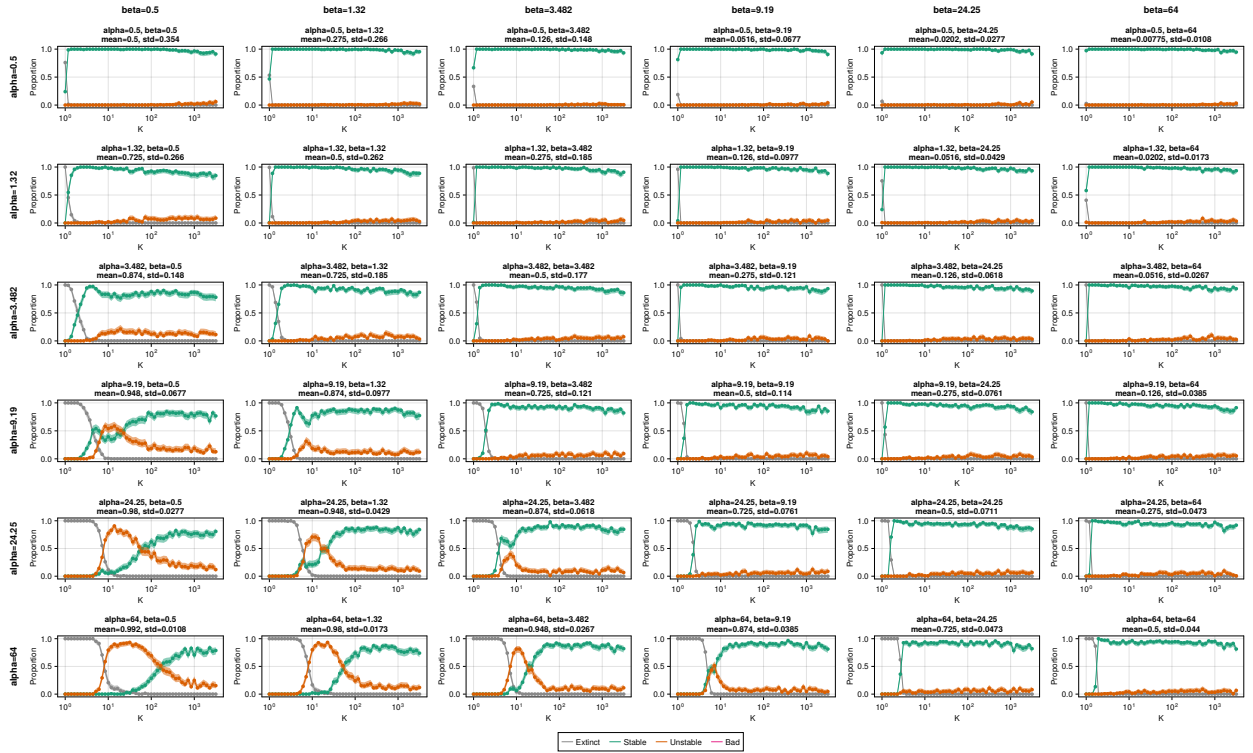


Figure 4: Proportion of extinct/stable/unstable outcomes vs. K , tiled by α (rows) and β (columns).

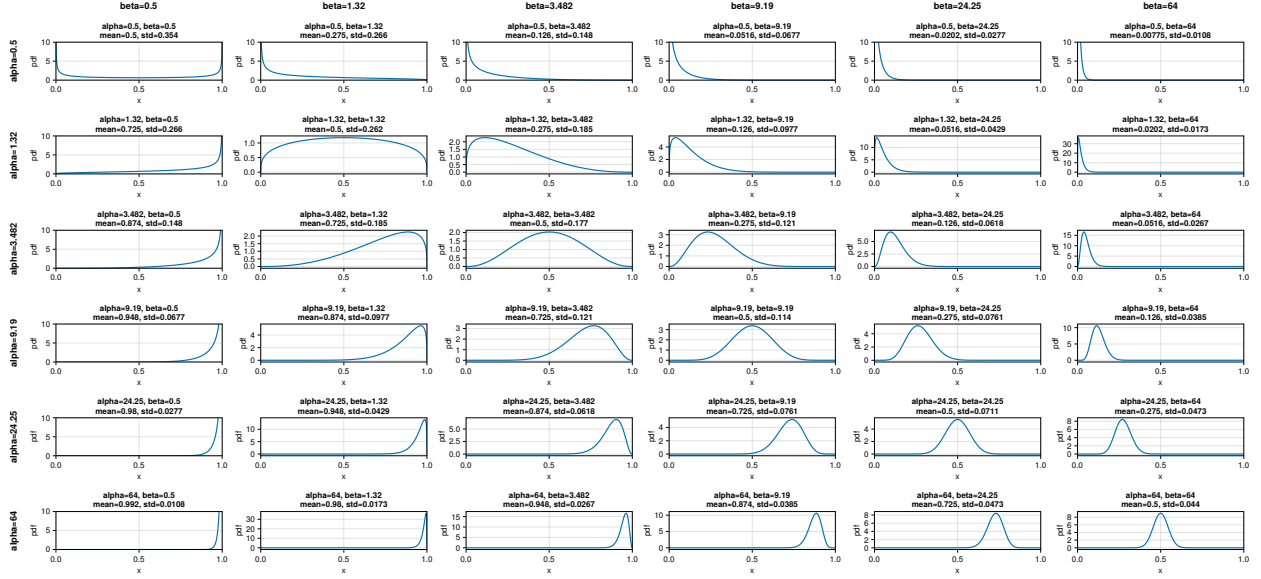


Figure 5: The sampled leakage distribution, $\text{Beta}(\alpha, \beta)$, at each grid position – directly comparable panel-for-panel with the outcome proportions above.